Privify

A Computer Vision Pipeline for GDPR-Compliant Face Anonymization

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Abstract Privify is a Python-based computer vision pipeline that detects and anonymizes faces in video footage to support GDPR compliance for small and medium enterprises operating CCTV systems. The system combines a fine-tuned YOLOv8n detector (mAP@50 = 0.937 on a single-class face dataset) with a configurable Gaussian blur anonymizer. This report describes the pipeline architecture, training methodology, empirical evaluation, and the domain-shift failure mode observed when deploying the in-distribution-optimized detector on street-level CCTV scenes. This is a concrete illustration of the metric-vs-behavior gap in applied ML.

Contents

1 Problem Statement

The widespread adoption of video surveillance among Italian small and medium enterprises (SMEs) has created a regulatory paradox: the same technology that protects people and assets exposes their data to significant compliance risks under the General Data Protection Regulation (GDPR, EU 2016/679). A CCTV camera in a retail store, an office, or a logistics site daily captures dozens of recognizable faces (employees, customers, visitors) that constitute personal data under Article 4 of the GDPR. It is worth specifying that CCTV footage does not automatically qualify as biometric data protected under Article 9: this category applies only when images are processed through facial recognition systems for the unique identification of a natural person. However, even within the standard personal data regime, the data controller is required to adopt technical and organizational measures proportionate to risk, in accordance with the principles of accountability (Art. 5.2) and privacy by design (Art. 25). When recordings must be retained beyond the ordinary term, shared with third parties (insurance companies, law enforcement, audits), or even partially published, these measures must concretely reduce the identifiability of the subjects captured. For the average Italian SME, which typically lacks a dedicated Data Protection Officer and a substantial IT budget, this requirement often translates into inaction: videos are deleted for safety or, worse, retained in violation, with exposure to the sanctions provided under Article 83.5 of the GDPR, up to 20 million euros or 4% of annual worldwide turnover.

Existing market solutions polarize into two segments that leave the intermediate space uncovered. On one side, solutions such as Brighter AI or Pimloc Secure Redact offer production-grade pipelines targeting the enterprise segment, typically with pricing that is not publicly listed or tied to consumption/volume models. This commercial orientation makes them inaccessible to IT consultants serving SMEs as an off-the-shelf purchase. On the other side, manual solutions based on professional video editors (Adobe Premiere, DaVinci Resolve) require expert operators and production time linear in the number of frames, economically prohibitive on realistic CCTV volumes. Numerous open-source face blurring projects also exist (typically educational notebooks or single-file scripts on GitHub), but they rarely offer the level of engineering rigor required for professional deployment: automated testing, continuous integration, explicit versioning of models and datasets, traceable architectural decisions. The opportunity space for IT consultants working with Italian SMEs is therefore a system that combines accessibility (open-source, self-hostable) with engineering maturity (auditable, tested, documented), characteristics that the GDPR itself rewards through the accountability principle of Article 5.2.

Privify is designed to occupy this space. It is an open-source computer vision pipeline that automatically detects faces in CCTV footage and applies a configurable Gaussian blur as a visual redaction measure. Whether this operation qualifies as anonymization in the strict GDPR sense (Recital 26) depends on contextual parameters such as blur intensity, frame resolution, and the presence of other identifiers in the scene. Recent research has demonstrated that Gaussian blur can be partially attacked through deblurring and re-identification techniques [citation placeholder: gaussian blur deanonymization paper]; the system is therefore designed to allow the data controller to calibrate parameters in an informed manner and to document the technical choice adopted. The system combines a YOLOv8n detector fine-tuned on a single-class face dataset (mAP@50 = 0.937 on the validation set) with a stateless OpenCV-based anonymizer. The entire pipeline is written in Python, accompanied by an automated test suite, continuous integration on GitHub Actions, and Architectural Decision Records documenting every technical choice. The target deployment is the IT consultant working with Italian SMEs: the system must run on a laptop or a small server without dedicated GPU, process archived video in times acceptable for batch use, and produce redacted output that the data controller can legitimately retain or share. The extension of the detector to the license plate class, part of Privify's original positioning for the Italian CCTV context (drive-through, private parking, commercial entrances), is planned as future work for v0.2.

This report documents the development of Privify in its first iteration (v0.1), from the initial architectural choice through the detector fine-tuning and the empirical evaluation on test videos. The main contribution is not algorithmic novelty: YOLOv8 and Gaussian blur are established techniques. It is instead the engineering quality of the integration and the transparency with which both the successes (high detection metrics in the in-distribution domain) and the limits (qualitative failure on street-level CCTV scenes, due to a textbook case of domain shift between training and deployment) are documented. The latter evidence, far from constituting a defect to conceal, represents the true didactic contribution of the project: a concrete demonstration of the gap between performance metric and qualitative behavior that characterizes applied machine learning, and an explicit roadmap to address it in subsequent iterations.

2 Methodology

TODO: pipeline overview, design decisions (Wrapper pattern, lazy loading, frozen dataclasses, DI), choice of YOLOv8n + Gaussian blur, training setup, dataset strategy. ~2 pagine.

2.1 Pipeline Architecture

2.2 Detection Model

2.3 Anonymization

2.4 Training Setup

3 Experimental Results

TODO: metriche del modello fine-tuned, training curves, qualitative evaluation, comparison vs COCO baseline. ~2 pagine.

3.1 Quantitative Metrics

3.2 Training Convergence

3.3 Qualitative Evaluation

4 Failure Analysis

TODO: domain shift documentato, root cause analysis, mitigation roadmap. ~1.5 pagine. SEZIONE FORTE — qui c'è il vero contributo narrativo del progetto.

4.1 Domain Shift on Street-Level CCTV

4.2 Root Cause Analysis

4.3 Mitigation Roadmap

5 Ethical Considerations

TODO: privacy by design, GDPR principle of minimization vs effectiveness, biometric quasi-identifiers, dataset licensing (CC BY 4.0), model licensing (AGPL-3.0), responsible deployment. ~1.5 pagine.

5.1 Privacy by Design

5.2 GDPR Alignment

5.3 Dataset and Model Licensing

5.4 Responsible Deployment Considerations

6 Conclusions

TODO: ricap, lessons learned, future work. ~0.5 pagine.

7 References

TODO: bibliografia minima (3-6 references). Citare YOLOv8 paper, McPherson 2016 sulla reversibilità di pixelation, GDPR text, Roboflow dataset attribution, eventuali altri paper letti.